



Catch Delinquency Before It Costs You

Early intervention for auto
lending, on one governed
lakehouse

Audience: Chief Risk Officer (sponsor) and Collections
& Risk Ops lead (owner)



The business outcome

- Predict which current loans will roll 30+ days past due NEXT cycle, and give collections a ranked queue to intervene before the roll.
- The result the business feels: lower net charge-offs and a more efficient collections team.
- No change to how loans are booked.



The problem, in your terms

- Collections is reactive: a loan is worked only after it is already delinquent, when cure is harder and costlier.
- Capacity is finite and today spread without prioritization.
- The cost shows up as early-stage roll rate, roll-to-charge-off, and net charge-offs (the KPIs the CRO reports).
- On this synthetic 5,000-loan book (~\$134.6M outstanding), 3.2% of current loans roll to 30+ DPD each cycle.



One integrated journey, not six silos

- Lakeflow ingests raw loan and payment data (bronze to silver with data-quality rules).
- Unity Catalog governs it (comments, governed tag policy, grants, lineage).
- ML and GenAI score every current loan and write a plain-language briefing.
- Lakebase serves the scored queue, the briefing, and operational cases (all also queryable in Unity Catalog).
- Genie answers questions in plain English, including over the live collection cases.
- A Lakebase-native Databricks App surfaces the queue, briefing, an embedded Genie assistant, and case actions.



Proof it works (from the actual build)

- Next-month 30+ DPD classifier, registered in Unity Catalog.
- ROC AUC 0.74: given a loan that rolls and one that does not, the model ranks the roller higher 74% of the time.
- Risk ranks monotonically with FICO, from 0.5% in the 780+ band up to 11.2% below 580.
- 3.7x lift in the top 5%: the model's highest-risk slice is ~3.7x more likely to actually roll than the average loan.
- Today's queue: 153 High-risk loans, \$4.4M exposure, surfaced with an AI briefing and recommended action.



Quantified impact (your KPIs)

- Collections hit rate: 11.8% on the worked queue vs 3.2% baseline, about 3.7x more productive contacts.
- Intervene on the 153 High-risk loans each cycle instead of spreading effort or waiting.
- Illustrative: curing 25% of the \$4.4M High-risk exposure at ~55% loss-given-default is ~\$0.6M in net charge-offs avoided per cycle.
- No new cost to book loans: this runs on data you already have.



Value for the executive sponsor (CRO)

- Lower net charge-offs and provisioning by shifting collections from reactive to predictive.
- One governed platform from raw data to app: auditable lineage, governed tag policies, and access control.
- Scales and compounds to pricing, line management, and recovery on infrastructure you already run.



Value for the collections & risk ops lead

- A ranked daily queue instead of a flat list: work the riskiest dollars first.
- An AI briefing that explains the queue and recommends an action in plain language.
- A Genie assistant embedded in the cockpit: ask in plain English, no SQL and no switching tools.
- Case workflow built in: open and assign a case from the same screen, written back to Lakebase and instantly queryable in Genie and Unity Catalog.



Why Databricks, why integrated matters

- The hand-offs between ingest, governance, ML, serving, natural language, and the app are the usual failure points.
- Here they are one platform, one security model, one copy of governed data.
- From a raw file to a business decision with no data movement to bolt-on tools: faster to value and cheaper to run.



Next steps

- Point this at a real (scrubbed) book and validate the lift on your charge-off history.
- Agree the intervention playbook (who gets called, what offer) for the High tier.
- Pilot with the collections team for one cycle; measure roll-rate and hit-rate deltas.
- Expand the governed lakehouse pattern to the next risk use case.
- All figures are from a synthetic dataset built for this prototype; dollar impact is illustrative and to be validated against your actuals.

